Design reference and requirements for AI product Yue Xin

AI toys fall into two categories: pure voice interaction and the need for augmented visual MultiModal Machine Learning. The software design refers to the Espressif Audio-Video Development Framework.

ESP-ADF; Hardware design refers to the Espressif Voice Development Board ESP32-S3-Korvo-2 and ESP32-P4 Development Board; Related cloud and large

The selection of the model solution can be contacted with Yue Xin for communication and understanding. Korvo development board is recommended

Selection of chips and modules

1. For most AI toys or early education hardware categories, Espressif ESP32-S3 can be selected by default for development
   - Chip selection recommendations are shown in the following table
   - Regarding MIC selection, a single MIC is recommended by default; if far-field voice wake-up is required, choose a dual MIC
     - Single MIC can also perform full-duplex RTC intercom
     - It should be noted that the selected external Codec must support hardware loopback (the loopback reference signal is used for the AEC algorithm), similar to ES8311
     - For toy categories, if the wake-up distance is only required to be within 1m, a single MIC can also be selected
   - If the COB form is not used, the corresponding module can be selected as ESP32-S3-WROOM-1
     - For those who are new to development or have relatively limited RF experience, it is more recommended to choose the module, and COB can be considered for subsequent hardware upgrades.
2. If there is a requirement for H264 video or the screen resolution is ≥480P, you can choose Espressif ESP32-P4 for development
   - For example, in addition to voice interaction, subsequent toy forms need to add video interaction capabilities such as visual intercom;

- Or add a visual model, increase Camera access, and perform MultiModal Machine Learning model recognition

模型	MIC 数量	唤醒需求	NS/AEC	屏幕需求	选型建议
语音	单	按键	无	无	S3 或 C3
语音	单	按键	需要	无	S3R2
语音	单	按键	NA	≤ 320P	S3R2
语音	双/单	语音	需要	无	S3R8
语音	双	语音	需要	≤ 480P	S3R8
视觉	单	按键	需要	无	S3R8/P4
视觉	单	NA	NA	480P 或以上	P4

The AEC algorithm mainly operates on full-duplex interaction and voice wake-up interruption operations

If the default DOA sound source algorithm is required, it will use 2MIC + S3R8 for processing

Regarding voice wake-up, if there are cost requirements, an external MCU (voice wake-up chip) such as ESP32-C3 can also be used for processing, but development

and the experience will be slightly inferior to ESP32-S3

Regarding the vision model, ESP32-P4 is recommended by default. If the functionality is not complex, such as photo recognition, ESP32-S3R8 can also be used.

Send

Requirements for MIC and Peripheral Selection

1. For the single MIC solution, it does not involve array requirements and has relatively low requirements
   - MIC can choose a high-sensitivity analog microphone, which has a lower cost; the sensitivity is recommended to be no less than -38 dB, and the signal-to-noise ratio no less than 60 dB
   - The peripheral Codec can be selected to be similar to ES8311, which has single input and single output and comes with an echo reference signal
2. For the dual MIC solution
   - Involving speech algorithms, the requirements for MIC selection are slightly higher

    - Under 1 Pa sound pressure, the sensitivity of the analog microphone is not less than -38 dBV, and the sensitivity requirement of the digital microphone is not less than -26 dB
    - The tolerance is controlled within ±2 dB, and for microphone arrays, a tolerance of ±1 dB is recommended.
    - Signal-to-Noise Ratio (SNR) is not less than 62 dB, recommended > 64 dB
    - The fluctuation of the frequency response within the range of 50 ~ 16 kHz is within ±3dB.
    - PSRR > 55 dB (MEMS mic)
  - Microphone arrays also have certain design requirements
    - The phase difference between microphones in the microphone array is controlled within 10°.
    - For the structural design of each microphone in the microphone array, it is recommended to adopt the same design to ensure the consistency of the structural design.
    - The microphone spacing should be 4 - 6.5 cm, the axis connecting the two microphones should be parallel to the horizontal line, and the centers of the two microphones should be as close as possible to the center of the product in the horizontal direction.
    - The peripheral Codec needs to support interfaces with at least three inputs (2 channels of MIC + 1 channel of feedback), and some digital PAs with feedback or Codec combinations can be selected

Design requirements for echo reference signal

1. If it is not an integrated Codec (including internal hardware loopback), but rather requires DA loopback to AD, then the echo reference signal should be as close as possible

On the speaker side, it is recommended to sample after the DAC and before the PA pre-stage.

2. When the speaker volume is at maximum, the echo reference signal input to the microphone must not have saturation distortion, and the THD of the speaker amplifier output at maximum volume meets the requirements

100 Hz is less than 10%, 200 Hz is less than 6%, and frequencies above 350 Hz are less than 3%

3. When the speaker volume is at maximum, the sound pressure picked up by the microphone does not exceed 102 dB @ 1 kHz
4. The voltage of the echo reference signal does not exceed the maximum allowable input voltage of the ADC; if the voltage is too high, an attenuation circuit needs to be added
5. To extract the reference echo signal from the output of a Class D power amplifier, a low-pass filter needs to be added, and the recommended cutoff frequency of the filter is > 22 kHz
6. When playing at maximum volume, the peak value of the feedback signal is -3 to -5 dB.

Suggestions for the overall structure design of the machine

> The overall cavity requirements for single-MIC and multi-MIC array solutions are basically the same; if there is no AEC requirement, the requirements for the cavity structure can be relaxed.
>
> For visual models, only the selection and focal length of the Camera need to be noted

1. Requirements for the hardware cavity of MIC
   - The diameter or width of the microphone hole is recommended to be greater than 1 mm, the pickup tube should be as short as possible, and the resonant frequency of the microphone and structural components should be ensured to be above 9 kHz
   - The depth-to-diameter ratio of the pickup hole is less than 4:1, the recommended shell thickness is 1 mm, and if the shell is too thick, the opening area needs to be increased
   - A dust screen can be considered for installation on the microphone hole for protection
   - The microphone holes must not be blocked. For the bottom-mounted microphone holes, a protrusion should be added to the structure to prevent them from being blocked by the desktop or other objects.
   - Mike should be placed away from objects such as speakers that generate noise or vibration, and isolated and cushioned from the speaker sound cavity by rubber pads or the like
2. It is necessary to ensure the sealing of the entire machine cavity, i.e., cavity isolation, to ensure that MIC recording is not affected by internal feedback and reverberation from speaker playback.
   - Ensure that the speaker has a separate sound cavity for playback, and the microphone is completely isolated from the playback cavity; some air holes caused by structural components are isolated by means such as foam or gluing
   - The playback surface should be on a different plane from the MIC recording surface as much as possible
   - Silicone sleeves or foam can also be added between the microphone and the device housing for sealing and shock absorption

Suggestions on Hibernation and Wake-up Methods

> For AI toy products, whether or not they are full-duplex RTC conversations, battery-powered products are recommended to have a sleep mechanism and a wake-up mechanism, with constant power

Products do not need to

1. When there is no "effective recording" within 5/10 minutes after the conversation ends, the device will exit the conversation mode and the entire unit will enter DeepSleep or LightSleep mode to save power with low power consumption.
   - No interaction ≠ not recording. When the product is not in use, be sure to turn off the sound pickup system and enter the sleep state to save power and prevent accidental triggering.
   - Even when using the voice wake-up interaction mode, the power consumption of the whole machine in the voice wake-up waiting state > 50mA is also a significant burden for battery-powered products
   - You can use the button to force a shutdown or let the AVD detect + cloud determine whether to exit the interactive mode. After detecting no "valid recording", play a similar prompt tone like "Little friend, I'm going to take a break. Wake me up anytime you need xxx" to strengthen user communication stickiness.
2. When it is necessary to use, the system can be awakened through the "button"
   - button needs to be connected to RTC_IO, a type of low-power IO, to wake up the system
   - Waking up the essence is to some extent equivalent to system restart. Therefore, it is recommended to play a similar prompt tone like "Little friend, I'm here again xxxxx" immediately upon waking up to accommodate the system initialization time and reduce user perception at the same time.
   - After the system wakes up, proceed with button-based dialogue, RTC full-duplex dialogue, or voice wake-up interruptive dialogue
3. As for the size of the battery, it is defined according to the structure and requirements of the product

📌 Main chip sleep can achieve DeepSleep <10uA, LightSleep keep-alive DTIM10 < 700uA.

Some development verification methods

Microphone Array Consistency Verification Method

1. At 0.5 meters directly above the microphone, white noise is played, with the sound level at the microphone being 90 dB
2. Record for more than 10 seconds using a microphone array, check whether the recording amplitude and audio sampling rate of each mic are consistent, and ensure that the difference can be within 3 dB

within

Cavity airtightness verification method

1. The microphone pickup hole can be sealed with materials such as modeling clay. After sealing, the amplitude attenuation of the microphone's collected signal should be 25 dB to be considered qualified, with 30 dB recommended.

Test Method (Silent Environment):

2. At 0.5 meters directly above the microphone, white noise is played, with the sound level at the microphone being 90 dB
3. Record for more than 10 seconds using a microphone array and save it as audio file A
4. Seal the microphone pickup holes with materials such as modeling clay, record for more than 10 seconds using the microphone array, and save it as recording file B
5. When comparing the spectra of two files, it is necessary to ensure that the overall attenuation within the 100 ~ 8 kHz frequency band is more than 30 dB

AEC Effect and Recovery Reference Verification

1. can use algorithm project to obtain 4 channels of audio (2 channels of MIC + AFE feedback + audio after AEC cancellation) for the pre- and post-stage of AEC,

Then use Audacity to check whether there are issues such as saturation, data loss, and left-right channel misalignment in the aec_in audio